

Ans Sheet: CS & AI Master Exam - Rohan Gupta (Roll: CS-2026-003)

Q1: Backpropagation in Neural Nets

Backpropagation adjusts neural net weights using loss function.

It uses back propagation of errors from output layer back to input nodes.

If loss is high, weights are changed.

Q2: QuickSort and MergeSort

QuickSort is fast sorting algorithm with time complexity $O(N)$.

MergeSort splits array into halves and has time complexity $O(N \log N)$.

MergeSort uses extra memory space. QuickSort is faster for array sorting.

Q3: ACID Properties

A = Atomicity

C = Consistency

I = Isolation

D = Durability

Transactions in SQL database must be ACID compliant so database does not crash during power loss.

Q4: OOP Concepts

Classes and Objects are main OOP concepts.

Inheritance allows class to inherit methods.

Encapsulation wraps variables in class.

Q5: RSA Cryptography

RSA is security algorithm used in HTTPS and internet passwords.

It generates public and private key pairs.

Messages are encrypted with key so hackers cannot read packets.